

HOTEL PRICING & CUSTOMER SATISFACTION



WEB SCRAPING



Introduction

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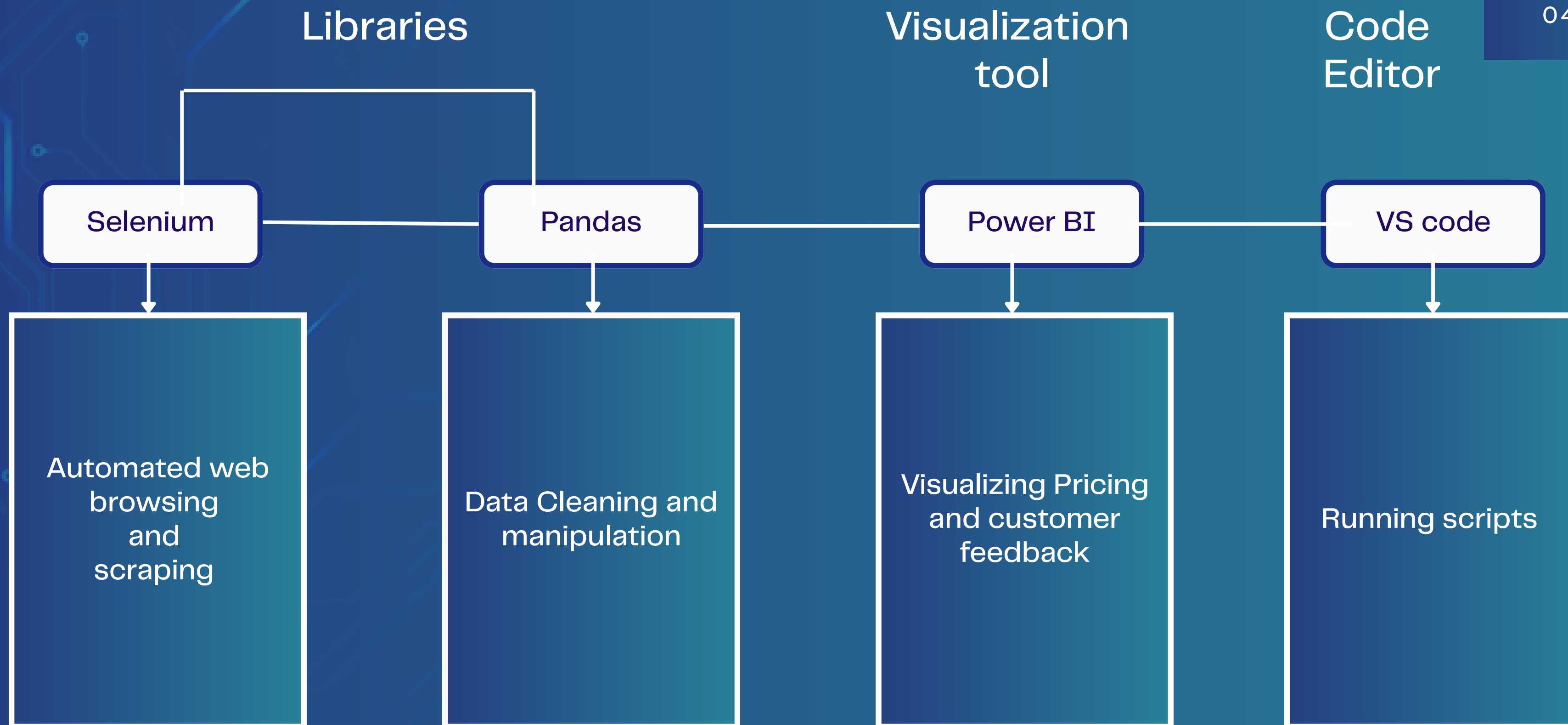
INTRODUCTION

This project focuses on developing a web scraper to analyze hotel prices and customer satisfaction on Booking.com. It covers five North African countries; Tunisia, Morocco, Algeria, Libya, and Mauritania and examines data across different seasons to identify pricing trends and satisfaction patterns. By using web scraping, the project aims to support better decision-making through customer feedback analysis, competition monitoring, and market insights. This will be done in 3 phases : web scraping, data cleaning and data visualization.

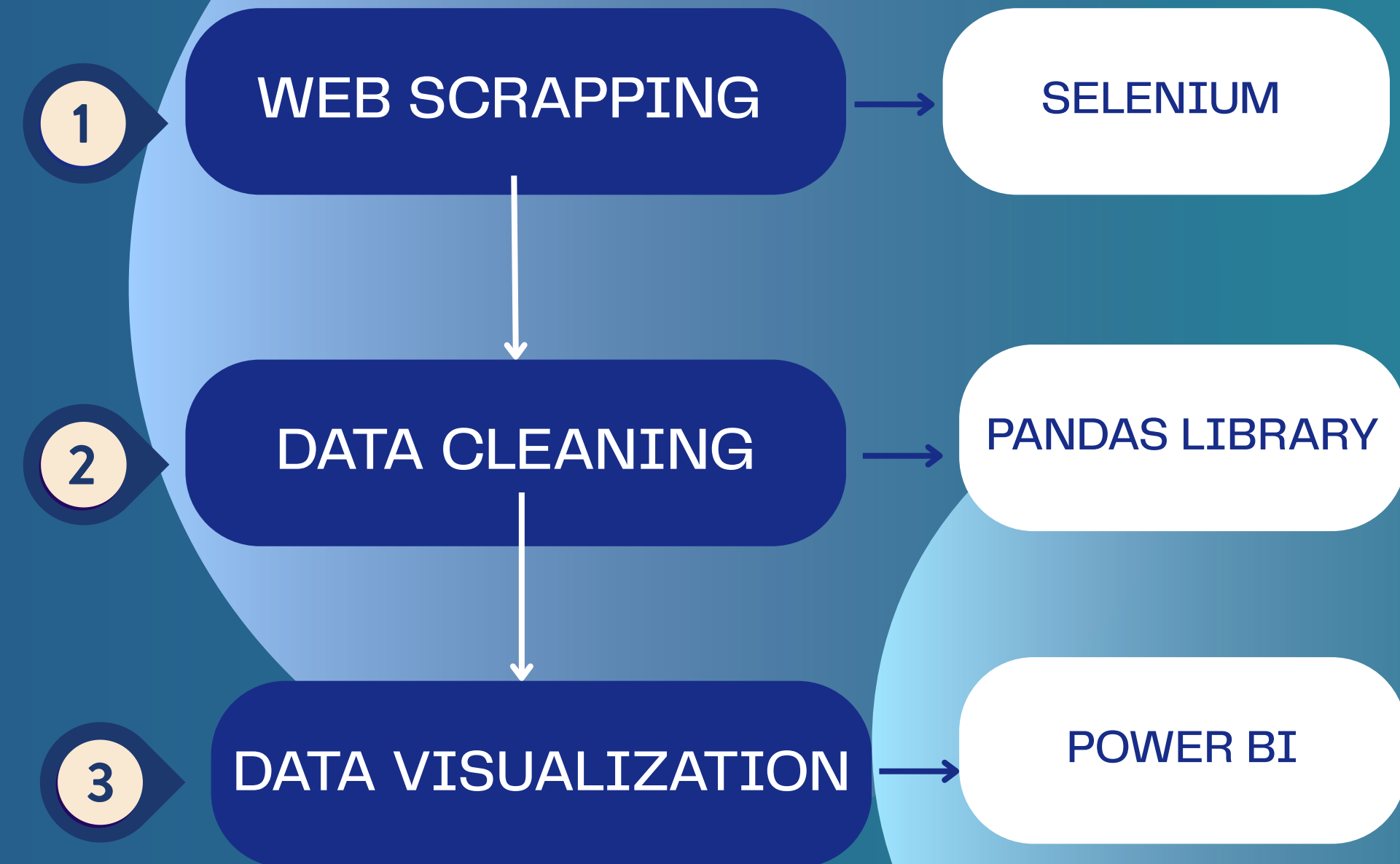


TOOLS & LIBRARIES





Our Framework consists of 3 core steps from the automation of the process of collecting customer feedback and hotels pricing to analyzing the satisfaction levels through graphical representation





This script automates the scraping of hotel data from Booking.com across North African countries using Selenium. It collects information such as hotel names, prices, ratings, and locations for different seasons, and saves the results into a structured CSV file for analysis.

PURPOSE OF THE SCRIPT





Scraping Tool

Selenium WebDriver with Chrome, using automated driver installation via webdriver-manager



Target Website

Booking.com hotel search results pages



Data Extracted

Hotel name, price, review score, review label, number of reviews ...



Output

One consolidated CSV file (north_africa_hotels.csv) containing all scraped data



Error Handling

Uses try-except blocks and retry mechanism (up to 3 attempts) to handle connection issues and missing elements



Data Collection Strategy

Multiple countries and cities with different seasonal date scenarios (Spring, Summer, Low Season)



Dynamic Loading Handling

Automated scrolling to load additional hotel listings



Progress Saving

Data is saved incrementally after each country-season combination to prevent data loss

07/19





```
4
5 # -----
6 # 0. LOAD
7 # -----
8 INPUT_PATH = r"C:\Users\ramez\Downloads\north_africa_hotels.csv"
9 OUTPUT_PATH = r"C:\Users\ramez\Downloads\north_africa_hotels_clean.csv"
10
11 df = pd.read_csv(INPUT_PATH)
12
13 print("=" * 60)
14 print(f"Loaded : {df.shape[0]} rows x {df.shape[1]} columns")
15 print("=" * 60)
16
17 # -----
18 # 1. STRIP WHITESPACE
19 # -----
20 str_cols = df.select_dtypes(include="object").columns
21 df[str_cols] = df[str_cols].apply(lambda col: col.str.strip())
22
23 print("\n[1] whitespace stripped")
24
25 # -----
26 # 2. DROP DUPLICATES
27 # -----
28 df.drop_duplicates(inplace=True)
29 df.reset_index(drop=True, inplace=True)
30
31 print("\n[2] duplicates removed")
```

DATA CLEANING



1

Load Data : Import raw scraped dataset

2

Remove Duplicates & Clean Text : Drop duplicate rows and strip whitespace

3

Fix Review Data : Extract numeric scores and standardize review labels

4

Convert Numeric Fields : Clean and convert price and review count into numeric format

5

Process Location : Extract and convert distance into kilometers

6

Handle Missing & Invalid Data : Remove or fix incomplete and inconsistent rows

7

Feature Engineering : Create stay_nights and stars_rating

8

Finalize Dataset : Rename columns, optimize data types, and save clean dataset



DATA VISUALIZATION



After collecting and cleaning hotel data from Booking.com, we used data visualization to uncover key pricing and customer satisfaction trends. Through interactive dashboards in Power BI, we highlighted patterns across seasons, cities, and countries, such as average prices, rating distributions, and review volumes, making it easier to analyze and interpret large-scale data.

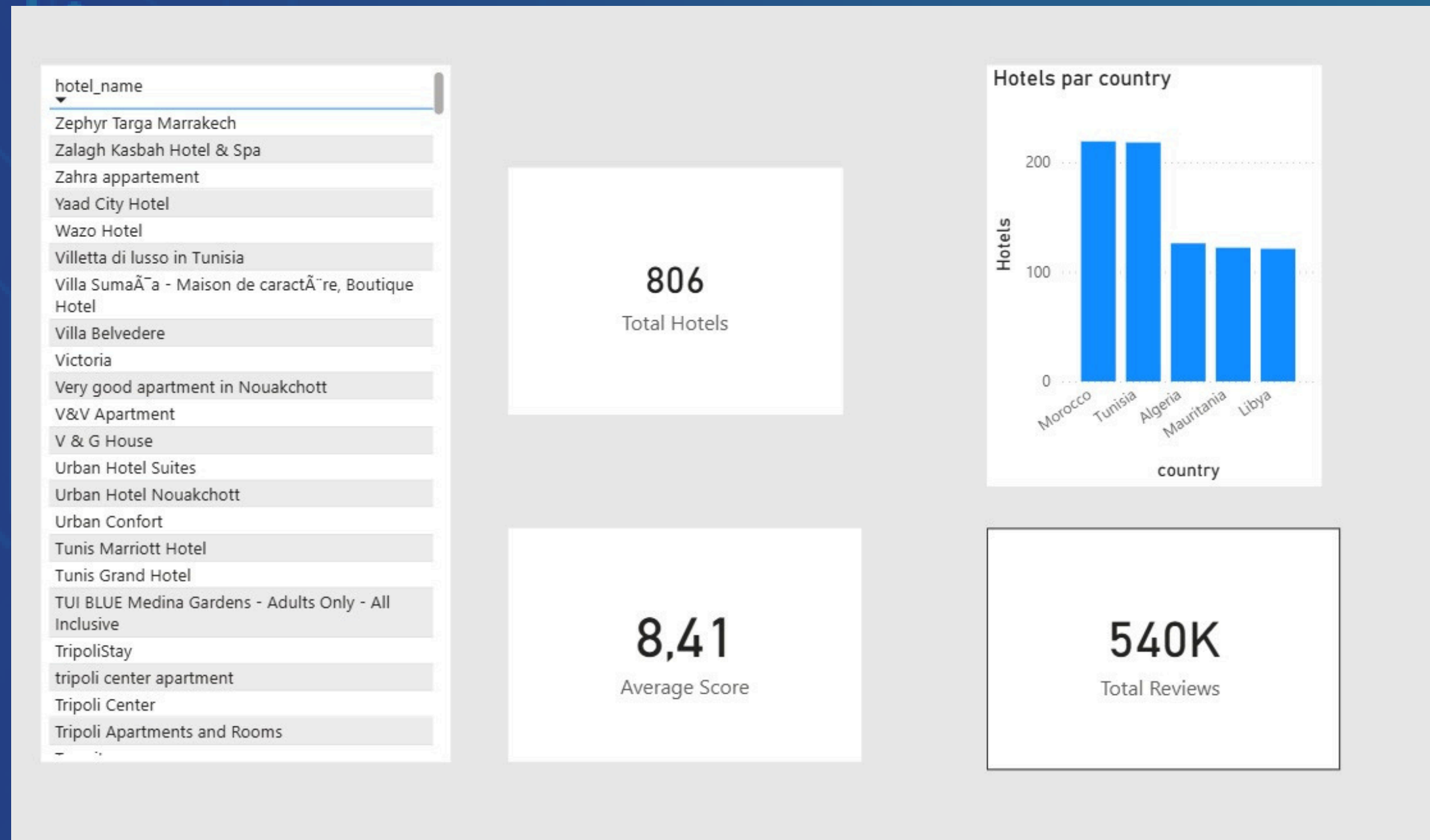


RESULTS & INTERPRETATIONS



PORTFOLIO OVERVIEW & HOTEL DISTRIBUTION BY COUNTRY

12/19



Total Hotels (KPI Card): The dataset covers 806 hotels in total across North Africa, providing a substantial and representative sample for regional analysis.

Average Score (KPI Card): The portfolio-wide average score sits at 8.41 out of 10, confirming that the overall quality of hotels across the region is consistently strong and falls within the Excellent tier.

Total Reviews (KPI Card): A combined 540K reviews underpin these scores, which gives the ratings strong statistical credibility — this is not a small or biased sample.

Figure-1





Hotels by Country (Bar Chart):

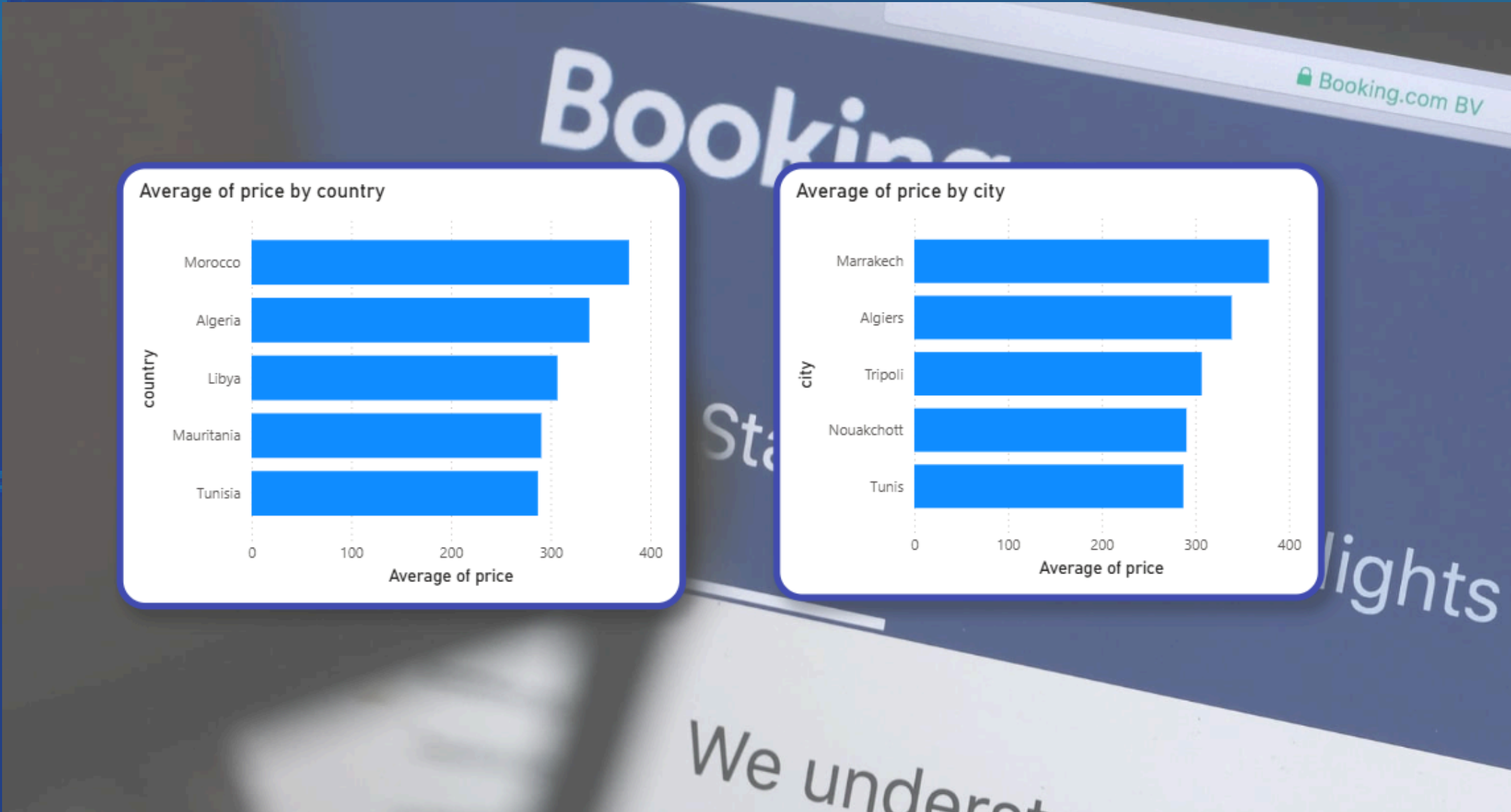
Morocco and Tunisia are the two most represented countries, both sitting just above 200 hotels each, making them the dominant markets in the dataset. Algeria follows in third place at roughly 150 hotels, while Mauritania and Libya are the least represented at around 130 hotels each. The relatively even spread across the last three countries suggests the dataset was curated to include regional diversity rather than reflecting actual market size — Libya in particular has far fewer real tourism listings than its bar implies.

Hotel Name Slicer (Left Panel):

The interactive slicer lists all 806 hotels alphabetically, allowing any single property to be isolated across all visuals. The variety of names visible — from large branded chains like Tunis Marriott and TUI BLUE to small apartments and boutique stays — confirms the dataset captures the full accommodation spectrum, from budget to luxury.



PRICE DISTRIBUTION ACROSS GEOGRAPHY & SEASON:



Average Price by Country : Morocco leads all countries with the highest average price (~370 TND), while Tunisia is consistently the most affordable despite being a major tourism hub. Algeria, Libya, and Mauritania sit in a tight mid-range band, suggesting relatively uniform pricing across those three markets.

Average Price by City : Marrakech is the most expensive city (~370 TND), consistent with Morocco's country-level lead. Tunis sits at the bottom (~295 TND), reinforcing Tunisia's budget-friendly positioning across the dataset.

Figure-2



Figure-3



Average Price by Hotel Name : A clear two-tier market exists. TUI BLUE Medina Gardens and Iberostar Waves Club Palmeraie sit between 800–870 TND — likely all-inclusive resorts — while most other named hotels fall well under 400 TND. Country-level averages are being pulled upward by a small number of premium outliers.

Average Price by Season : May (Spring) is the most expensive season (~340 TND), yet the gap between all three seasons is only about 40 TND. July, despite being peak summer, is actually the cheapest — indicating that hotels across the region apply very little dynamic seasonal pricing, which is a missed revenue opportunity



DISTANCE FROM DOWNTOWN VS. PRICE RELATIONSHIP

16/19



Overall Pattern : The vast majority of hotels cluster within 0–5 km of downtown, and within that zone prices range from near zero to over 700 TND. Being centrally located says very little about what a hotel charges.

No Clear Distance-Price Correlation Expensive hotels appear at nearly every distance band, including 15–30 km from downtown. These high-priced outliers far from city centers are almost certainly beachfront or resort properties, where amenities drive pricing more than urban proximity.

Key Takeaway : Budget hotels exist exclusively near city centers, while luxury pricing appears at all distances. Distance from downtown alone is not a reliable price predictor — the market needs to be segmented into city hotels vs. resort hotels before any meaningful pricing analysis can be done

Figure-4



REVIEW QUALITY, VOLUME & SCORE ANALYSIS:

17/19



Figure-5

KPI Cards The portfolio averages a strong **8.41 out of 10** across **806 total reviews**, placing it firmly in the Excellent tier and signalling broad guest satisfaction across the region





Review Distribution by Label (Pie Chart):

Excellent (36.6%) and Wonderful (33.1%) together account for nearly 70% of all reviews — a healthy, quality-skewed distribution. Exceptional is barely a sliver, meaning no hotel has yet broken through to consistently perfect guest experiences

Average Score by Country (Bar Chart):

Libya leads at ~8.6 and Morocco follows at ~8.5, both consistent with their premium market positioning. Algeria is the outlier — it is the second most expensive country yet scores the lowest at ~8.2, signalling a value-for-money gap that risks guest dissatisfaction.

Review Count vs. Score by Hotel (Scatter):

One hotel stands sharply apart with ~28K reviews and a score just above 5. This high-volume, low-quality outlier is likely a large budget property and is pulling down regional averages disproportionately

Average Score by Season (Line Chart):

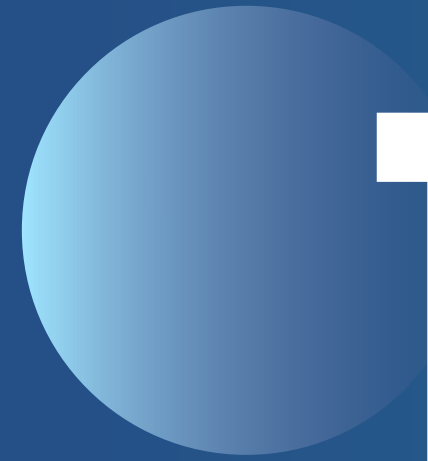
May (Spring) scores highest at 8.45, July dips slightly, and September (Low Season) records the lowest scores. Spring is the only season where both price and guest satisfaction peak simultaneously — making it the most strategically valuable period in the entire dataset.



CONCLUSION

This project demonstrates the effectiveness of web scraping and data visualization in analyzing hotel market trends. By automating data collection from Booking.com and interpreting it through interactive dashboards, we identified key insights on pricing and customer satisfaction across seasons and locations helping support better decision-making in the hospitality sector.





**THANK
YOU!**

